k — char p field

$\sigma : k \longrightarrow k$

$x \longmapsto x^p$

$$\begin{array}{c} T = \operatorname{Spec} B \\ \downarrow s \\ \operatorname{Spec} k \end{array}$$

Def (1) Frobenius twist $T^{(p)} := T \times_{\operatorname{Spec} k, \sigma} \operatorname{Spec} k = \operatorname{Spec}(B \otimes_{k,\sigma} k)$

i.e.
$$\begin{tikzcd} T^{(p)} \arrow[r] \arrow[d] & T \arrow[d, "s"] \\ \operatorname{Spec} k \arrow[r, "\operatorname{Spec}\sigma"] & \operatorname{Spec} k \end{tikzcd} \qquad \begin{tikzcd} k \arrow[r, "\sigma"] \arrow[d] & k \arrow[d] \\ B \arrow[r] & B \otimes_{k,\sigma} k \end{tikzcd}$$

(2) absolute Frobenius ; $\sigma : B \longrightarrow B$, $b \longmapsto b^p$ $\rightsquigarrow$ $\operatorname{Frob}_T : \operatorname{Spec} B \longrightarrow \operatorname{Spec} B$

(for non-affine schemes, taking p^{th} power on sections defines a morphism of sheaves)

(3) relative Frobenius ;
$$\begin{tikzcd} T \arrow[dr, dashed, "\exists !"] \arrow[drr, bend left, "\operatorname{Frob}_T"] \arrow[ddr, bend right, "s"'] & & \\ & T^{(p)} \arrow[r] \arrow[d] & T \arrow[d] \\ & \operatorname{Spec} k \arrow[r, "\operatorname{Spec}\sigma"] & \operatorname{Spec} k \end{tikzcd}$$

call $T \dashrightarrow T^{(p)}$ the relative Frobenius φ_T.

(4) $r \geq 1$. By induction, p^r-Frobenius twist is $T^{(p^r)} := \left(T^{(p^{r-1})}\right)^{(p)}$

relative p^r-Frobenius is $\varphi_T^r := \varphi_{T^{(p^{r-1})}} \circ \varphi_T^{r-1}$.

Concretely, $T^{(p^r)} \cong T \times_{\operatorname{Spec} k, \sigma^r} \operatorname{Spec} k = \operatorname{Spec}(B \otimes_{k,\sigma^r} k) =: \operatorname{Spec}(B^{(p^r)})$.

$\sigma^r : B \longrightarrow B$, $b \longmapsto b^{p^r}$ $\rightsquigarrow$ $\operatorname{Frob}_T^r : T \longrightarrow T$

$$\varphi_T^r = (\operatorname{Frob}_T^r, s) : T \longrightarrow T^{(p^r)}$$
$$\uparrow$$
$$B^{(p^r)} \longrightarrow B$$
$$b \otimes x \longmapsto x \cdot b^{p^r}.$$

ex. $B = k[t_1, \dots t_n]/(f_1, \dots f_m)$.

- $B^{(p^r)} \cong k[t_1, \dots t_n]/(f_1^{(p^r)} \dots f_m^{(p^r)})$ where $f = \sum_\alpha c_\alpha t^\alpha \rightsquigarrow f^{(p^r)} = \sum_\alpha c_\alpha^{p^r} t^\alpha$.

follows from $k[t_1, \dots t_n] \longrightarrow k[t_1, \dots t_n] \otimes_{k,\sigma^r} k$

$\sum_\alpha c_\alpha t^\alpha \longmapsto \sum_\alpha c_\alpha t^\alpha \otimes 1 = \sum_\alpha t^\alpha \otimes c_\alpha^{p^r} \longleftarrow \sum_\alpha c_\alpha^{p^r} t_\alpha \in k[t_1, \dots t_n]$.

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Formal consequences of the properties of fibre product: $\begin{smallmatrix} & U \\ & \downarrow \\ \mathrm{Spec}\, k & \end{smallmatrix}$.

- $(T \times_{\mathrm{Spec}\, k} U)^{(p)} \cong T^{(p)} \times_{\mathrm{Spec}\, k} U^{(p)}$
- $\varphi_{T \times_{\mathrm{Spec}\, k} U} = (\varphi_T, \varphi_U)$.
- $\begin{tikzcd} T \arrow[r, "f"] \arrow[d] & U \arrow[dl] \\ \mathrm{Spec}\, k & \end{tikzcd} \rightsquigarrow \begin{tikzcd} T \arrow[r, "\varphi_T"] \arrow[d] & T^{(p)} \arrow[d, dashed] \\ U \arrow[r] & U^{(p)} \end{tikzcd}$ call $f^{(p)} : T^{(p)} \dashrightarrow U^{(p)}$ the Frobenius twist of f.
- $(\varphi_T^{\,r})^{(p^s)} = \varphi_{T^{(p^s)}}^{\,r}$

in particular
- G finite flat$/k$ $\Rightarrow$ $G^{(p)}$ finite flat$/k$.
- φ_G group scheme hom.
- $(G^{(p)})^\vee \cong (G^\vee)^{(p)}$, hence $G^{(p)} \cong ((G^\vee)^{(p)})^\vee$.
- $f : G \to H$ group hom $\rightsquigarrow$ $f^{(p)} : G^{(p)} \to H^{(p)}$ group hom. and

$$\begin{tikzcd} G \arrow[r, "\varphi_G"] \arrow[d, "f"'] & G^{(p)} \arrow[d, "f^{(p)}"] \\ H \arrow[r, "\varphi_H"] & H^{(p)} \end{tikzcd}$$

- $f : G \hookrightarrow H \Rightarrow f^{(p)} : G^{(p)} \hookrightarrow H^{(p)}$
- $f : G \xrightarrow{\sim} H \Rightarrow f^{(p)} : G^{(p)} \xrightarrow{\sim} H^{(p)}$.

Def. (Verschiebung) $\varphi_{G^\vee} : G^\vee \longrightarrow (G^\vee)^{(p)} \overset{\vee}{\rightsquigarrow} \psi_G : \underset{\substack{\cong \\ G^{(p)}}}{((G^\vee)^{(p)})^\vee} \longrightarrow \underset{\substack{\cong \\ G}}{(G^\vee)^\vee}$

$$\psi_G^{\,r} = \psi_G^{\,r-1} \circ \psi_{G^{(p^{r-1})}} : G^{(p^r)} \longrightarrow G.$$

Lem. G, H finite flat$/k$.

(a) $\psi_{G \times_{\mathrm{Spec}\, k} H} = (\psi_G, \psi_H)$ via $(G \times_{\mathrm{Spec}\, k} H)^{(p)} \cong G^{(p)} \times_{\mathrm{Spec}\, k} H^{(p)}$.

(b) $G \xrightarrow{f} H \rightsquigarrow f^{(p)} : G^{(p)} \longrightarrow H^{(p)}$ s.t.

$$\begin{tikzcd} G \arrow[d, "f"'] & G^{(p)} \arrow[l, "\psi_G"'] \arrow[d, "f^{(p)}"] \\ H & H^{(p)} \arrow[l, "\psi_H"'] \end{tikzcd}$$

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Prop. (1) $\varphi_{\alpha_p} = 0 = \psi_{\alpha_p}$.

(2) $\varphi_{\mu_p} = 0$, $\psi_{\mu_p} = \mathrm{id}_{\mu_p}$.

(3) $\varphi_{\underline{\mathbb{Z}/p\mathbb{Z}}} = \mathrm{id}_{\underline{\mathbb{Z}/p\mathbb{Z}}}$, $\psi_{\underline{\mathbb{Z}/p\mathbb{Z}}} = 0$.

Proof. (1) recall $\alpha_p = \mathrm{Spec}\ k[t]/(t^p)$, so φ_{α_p} maps t to $t^p = 0$.

since $\alpha_p^\vee \cong \alpha_p$, it follows that $\psi_{\alpha_p} = 0$.

(2) similarly $\mu_p = \mathrm{Spec}\ k[t]/(t^p - 1)$ so $\varphi_{\mu_p}(t) = t^p = 1$, which is the trivial homomorphism. ψ_{μ_p} will follow from (3), as $\mu_p^\vee = \underline{\mathbb{Z}/p\mathbb{Z}}$.

(3) recall $\underline{\mathbb{Z}/p\mathbb{Z}} = \mathrm{Spec}\left(\prod_{\mathbb{Z}/p\mathbb{Z}} k\right)$. $\varphi_{\underline{\mathbb{Z}/p\mathbb{Z}}}\left(\sum_{i\in\mathbb{Z}/p\mathbb{Z}} c_i e_i\right) = \sum_{i\in\mathbb{Z}/p\mathbb{Z}} c_i e_i^p = \sum_{i\in\mathbb{Z}/p\mathbb{Z}} c_i e_i$,

as $e_i = (0, \dots, \underset{i^{\text{th}}}{1}, \dots, 0)$. So $\varphi_{\underline{\mathbb{Z}/p\mathbb{Z}}} = \mathrm{id}$, and $\psi_{\underline{\mathbb{Z}/p\mathbb{Z}}}$ follows from (2). $\square$

Prop. $G = \mathrm{Spec}\ A$ finite flat $/k$. $S_n \circlearrowright A^{\otimes n}$ by permuting pure tensor factors.

(1) $\exists$ k-alg hom $\gamma : (A^{\otimes n})^{S_n} \longrightarrow A^{(p)}$ s.t.

(i) $\gamma(a^{\otimes p}) = a \otimes 1$

(ii) for $v = a_1 \otimes \cdots \otimes a_n$ with some $a_i \neq a_j$,

$$\gamma\left(\sum_{\sigma \in S_n/\mathrm{Stab}(v)} \sigma(v)\right) = 0.$$

(2) $\psi_G \rightsquigarrow \psi_A : A \longrightarrow A^{(p)}$ fits into

$$\begin{tikzcd}
A \arrow[r] \arrow[rr, bend left, "\psi_A"] \arrow[dr, bend right] & (A^{\otimes p})^{S_n} \arrow[r, "\gamma"] \arrow[d, hook] & A^{(p)} \\
& A^{\otimes p} &
\end{tikzcd}$$

comultiplication of G.

Proof. as k-algebras, we have natural

$$A \cong (A^\vee)^\vee, \quad (\mathrm{Sym}^p A^\vee)^\vee \cong (A^{\otimes p})^{S_n}, \quad A^{(p)} \cong ((A^\vee)^{(p)})^\vee.$$

also k char $p \Rightarrow (f_1 + f_2)^{\otimes p} = f_1^{\otimes p} + f_2^{\otimes p} \in \mathrm{Sym}^p A^\vee$. So

(1) $\exists!\ \theta : (A^\vee)^{(p)} \longrightarrow \mathrm{Sym}^p A^\vee$

$$f \otimes c \longmapsto c \cdot f^{\otimes p}$$

$\gamma := \theta^\vee$. (i) for $a \in A \longleftrightarrow e_a \in (A^\vee)^\vee$ and $f \otimes c \in (A^\vee)^{(p)}$,

$$\gamma(a^{\otimes p})(f \otimes c) = e_a^{\otimes p}(c \cdot f^{\otimes p}) = c \cdot f(a)^{\otimes p} = f(a) \otimes c = (e_a \otimes 1)(f \otimes c).$$

(ii) $\gamma\left(\sum_{\sigma \in S_n/\mathrm{Stab}(v)} \bigotimes_{i=1}^p a_{\sigma(i)}\right)(f \otimes c) = \sum_\sigma \left(\bigotimes_{i=1}^p e_{a_{\sigma(i)}}\right)(c \cdot f^{\otimes p}) = c \sum_\sigma \prod_{i=1}^p f(a_i)$, which is 0 as $p \mid \#\sigma$

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(2) By construction

$$\begin{tikzcd} (A^\vee)^{(p)} \arrow[r, "\theta"] \arrow[rr, bend left, "\varphi_{A^\vee}"] & \mathrm{Sym}^p A^\vee \arrow[r] & A^\vee \\ & (A^\vee)^{\otimes p} \arrow[u, two heads] \arrow[ur] & \end{tikzcd}$$

$\otimes f_i \mapsto \prod f_i$ (multiplication)

$\wr\vee$

$A \longrightarrow A^{\otimes p}$ comultiplication.

So the diagram dualizes to (2). $\square$

Prop. $G = \mathrm{Spec}\, A$ finite flat$/k$. Then $\psi_G \circ \varphi_G = [p]_G$ and $\varphi_G \circ \psi_G = [p]_{G^{(p)}}$.

Proof. The digram from prev. prop extends to

$$\begin{tikzcd} A \arrow[r] \arrow[rr, bend left, "\psi_A"] \arrow[dr] & (A^{\otimes p})^{S_n} \arrow[r, "\gamma"] \arrow[d, hook] & A^{(p)} \arrow[d, "\varphi_A"] \\ & A^{\otimes p} \arrow[r] & A \end{tikzcd}$$

comultiplication of G. multiplication

$\xrightarrow{\mathrm{Spec}}$

$$\begin{tikzcd} G & & G^{(p)} \arrow[ll, "\psi_G"'] \\ & G^{\times p} \arrow[ul, "\text{multiplication}"] & \\ & G \arrow[u, "\varphi_G"] \arrow[l] & \end{tikzcd}$$

$(g \cdots g) \leftarrow\!\shortmid g$

So $\psi_G \circ \varphi_G = [p]_G$.

Moreover $\varphi_G^{(p)} = \varphi_{G^{(p)}}$ so

$$\begin{tikzcd} G^{(p)} \arrow[r, "\varphi_{G^{(p)}}"] \arrow[d, "\psi_G"] & G^{(p^2)} \arrow[d, "\psi_{G^{(p)}}"] \\ G \arrow[r, "\varphi_G"] & G^{(p)} \end{tikzcd}$$

$\Rightarrow \varphi_G \circ \psi_G = \psi_{G^{(p)}} \circ \varphi_{G^{(p)}}$

$= [p]_{G^{(p)}}$. $\square$

Lem. $G = \mathrm{Spec}\, A$ finite flat$/k$. (1) φ_G iso $\Longleftrightarrow$ injective

(2) G cnctd $\Rightarrow$ A artinian local k-alg whose max ideal is the augmentation ideal of G

↳ recall: $I = \ker \epsilon$ ↳ counit.

Proof. (1) recall from last time $\ker \varphi_G \hookrightarrow G$.

then Grothendieck's theorem $\Rightarrow$ $0 \to \ker \varphi_G \to G \to \overset{=1}{G/\ker \varphi_G} \to 0$.

Now if $\ker \varphi_G = 0$ then $G \cong G/\ker \varphi_G$.

(2) also last time: G cnctd $\Longleftrightarrow$ A finite local k-alg.

$\overset{\text{alg}}{\Longrightarrow}$ finite k-algebra is artinian.

$A/I \cong \mathrm{im}(\epsilon)$ and $\mathrm{im}(\epsilon) = k$ so I is the max idl. $\square$

Prop. (will not prove, appending past results)

(1) G cnctd $\Longleftrightarrow$ φ_G^r vanishes, some r (can dualize for ψ criterion)

(2) G étale $\Longleftrightarrow$ φ_G iso.

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Prop. $G = \operatorname{Spec} A$ finite flat$/k$.

(1) $\ker(\varphi_G) = \operatorname{Spec}(A/I^{(p)})$. $\quad I^{(p)}$ is the ideal gen. by a^p, $a \in I$.

(2) if $\varphi_G = 0$ then $\dim G = p^{\dim_k I/I^2}$.

Proof.

Prop. G finite flat$/k$, cnctd. Then $\operatorname{ord}(G) = p^r$. (Some r)

Prop. $G = \operatorname{Spec} A$ finite flat$/k$, unit section e. Then $t_{G,e} \cong \operatorname{Hom}_{k\text{-gp}}(G^\vee, \mathbb{G}_a)$.

Prop. G ff$/k$ étale $\Leftrightarrow \operatorname{Hom}_{k\text{-gp}}(G^\vee, \mathbb{G}_a)$ vanishes

Thm. $k = \bar{k}$. (1) simple ff k-gp is either étale or cnctd.

(2) simple finite étale k-gps are $\underline{\mathbb{Z}/\ell\mathbb{Z}}$, ℓ prime.

(3) simple cnctd ff k-gps are μ_p, α_p.

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Frobenius endomorphism

- k char p.
- $\sigma : k \to k$, $x \mapsto x^p$.
- $T = \operatorname{Spec} B \to \operatorname{Spec} k$, $\begin{pmatrix} B \\ \uparrow \\ k \end{pmatrix} \rightsquigarrow \sigma_B : B \to B$, $b \mapsto b^p$
- $r > 0$.

$$\begin{tikzcd}
k \arrow[r, "\sigma^r"] \arrow[d] & k \arrow[d] \arrow[ddr, bend left] & \\
B \arrow[r] \arrow[drr, bend right, "\sigma_B^r"'] & B \otimes_k k =: B^{(p^r)} \arrow[dr, dashed, "\varphi_B^{[r]}"] & \\
& & B
\end{tikzcd}$$

$$\begin{tikzcd}
T \arrow[dr, dashed, "\varphi_T^{[r]}"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& T^{(p^r)} := T \times_{\operatorname{Spec} k} \operatorname{Spec} k \arrow[r] \arrow[d, "\cancel{\ }"] & \operatorname{Spec} k \arrow[d] \\
& T \arrow[r] & \operatorname{Spec} k
\end{tikzcd}$$

← if k perfect, this is iso. hence $\cancel{\ }$ is iso.

don't even need T affine.

$T \to \operatorname{Spec} k \Rightarrow$ local sections of T are char p, so $T(U) \to T(U)$, $s \mapsto s^p$ is morphism of sheaves.

+ id on spaces $\to$ abs. Frob. $T \to T$.

Niceties

$f : Y \hookrightarrow X$, $\operatorname{Spec} k$ — dim + finite-type. don't actually need here.

$$\begin{tikzcd}
Y \arrow[r, "\text{abs}"] \arrow[d, "\text{rel}"'] & Y \arrow[r, "f"] & X \arrow[dr, "\text{rel}"] & \\
Y' \arrow[rr, "p^r"'] & & & X'
\end{tikzcd}$$

$$\begin{tikzcd}
Y \arrow[r, "f"] \arrow[d, "\text{rel}"'] \arrow[dr, phantom, "\rightsquigarrow"] & X \arrow[d, "\text{rel}"] \\
Y' \arrow[r] & X'
\end{tikzcd}$$

"p^r-Frob. twist of f"

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Nice properties:

- $(T \times_k U)^{(p^r)} \simeq T^{(p^r)} \times_k U^{(p^r)}$ natural

  s.t. $\varphi^{[r]}_{(T \times_k U)} = \varphi_T^{[r]} \times_k \varphi_U^{[r]}$

- G aff k-gp $\rightsquigarrow$ $G^{p^{(r)}}$ natural aff k-gp.

- $\varphi_G^{[r]}$ k-gp hom

- G finite flat $\Rightarrow$ $G^{(p^r)}$ finite flat, $(G^{(p^r)})^\vee \cong (G^\vee)^{(p^r)}$.

- $f: G \to H$ k-aff gp hom.

  $\rightsquigarrow f^{(p^r)}$ k-gp hom.

  $\rightsquigarrow f$ clim $\Rightarrow f^{(p^r)}$ clim

  $\rightsquigarrow f$ iso $\Rightarrow f^{(p^r)}$ iso

$\psi_G = \varphi_{G^\vee}^\vee$ dual of rel F of $G^\vee$.

$\psi_G \circ \varphi_G = [p]_G$, $\varphi_G \circ \psi_G = [p]_{G^{(p)}}$

G cnnctd $\Leftrightarrow$ φ_G^r vanishes for some r

G étale $\Leftrightarrow$ φ_G iso

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